

VitaSpec_huiles

ME

2024-05-10

Huiles achetées dans différents pays d'Afrique de l'Ouest en 2024. Analysées par Amel Joualli

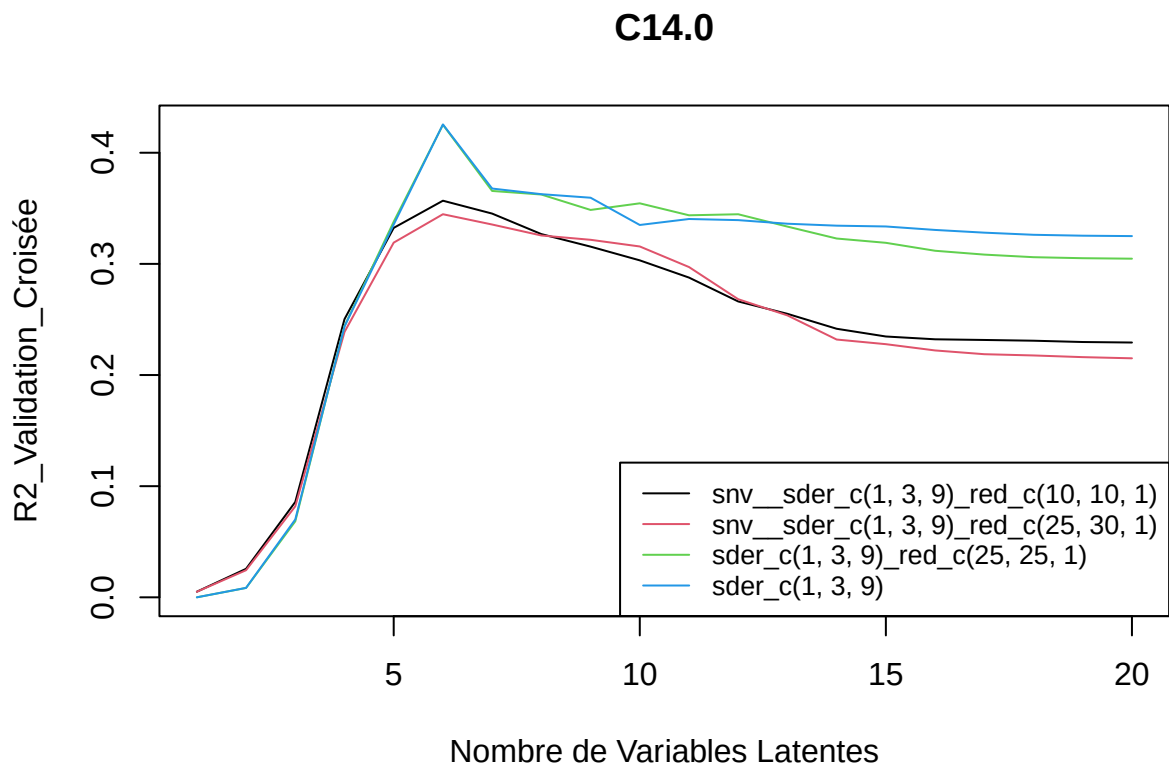
Dans les spectres, manquent les échantillons 34 et 87

Mise en forme des données et save

Chargement des prétraitements

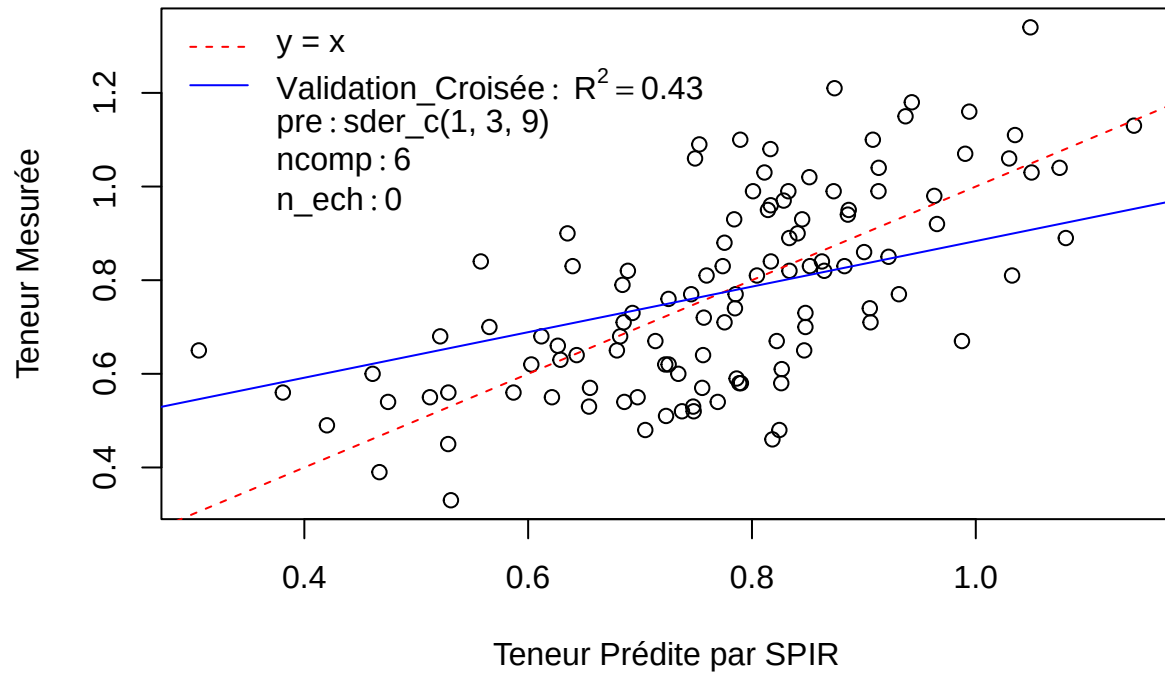
Calibrations sur spectres moyen

[1] "C14.0"



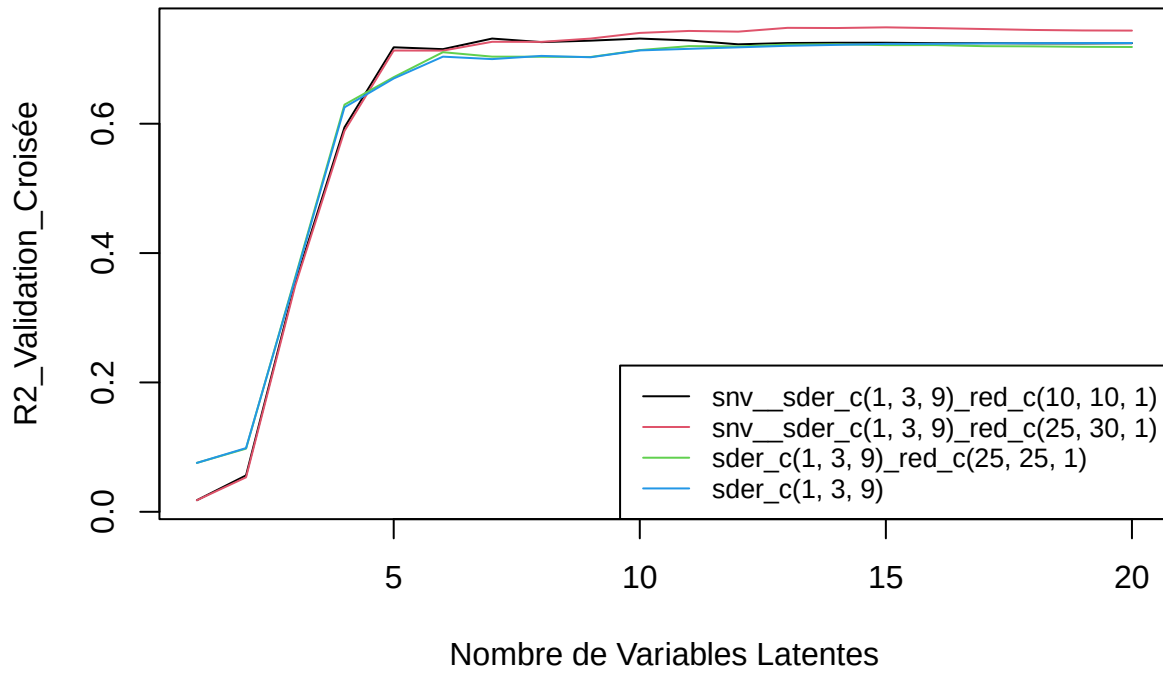
```
## [1] "sder_c(1, 3, 9)"
## [1] "R2"
## [1] 0.00 0.01 0.07 0.24 0.34 0.43 0.37 0.36 0.36 0.34 0.34 0.34 0.34 0.33 0.33 0.33 0.33 0.33 0.33 0.33 0.33
## [1] "SEP"
## y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1
## 0.21 0.21 0.20 0.18 0.17 0.16 0.17 0.17 0.17 0.17 0.18 0.18 0.18 0.18 0.18 0.18 0.18 0.18 0.18 0.18 0.18
```

C14.0



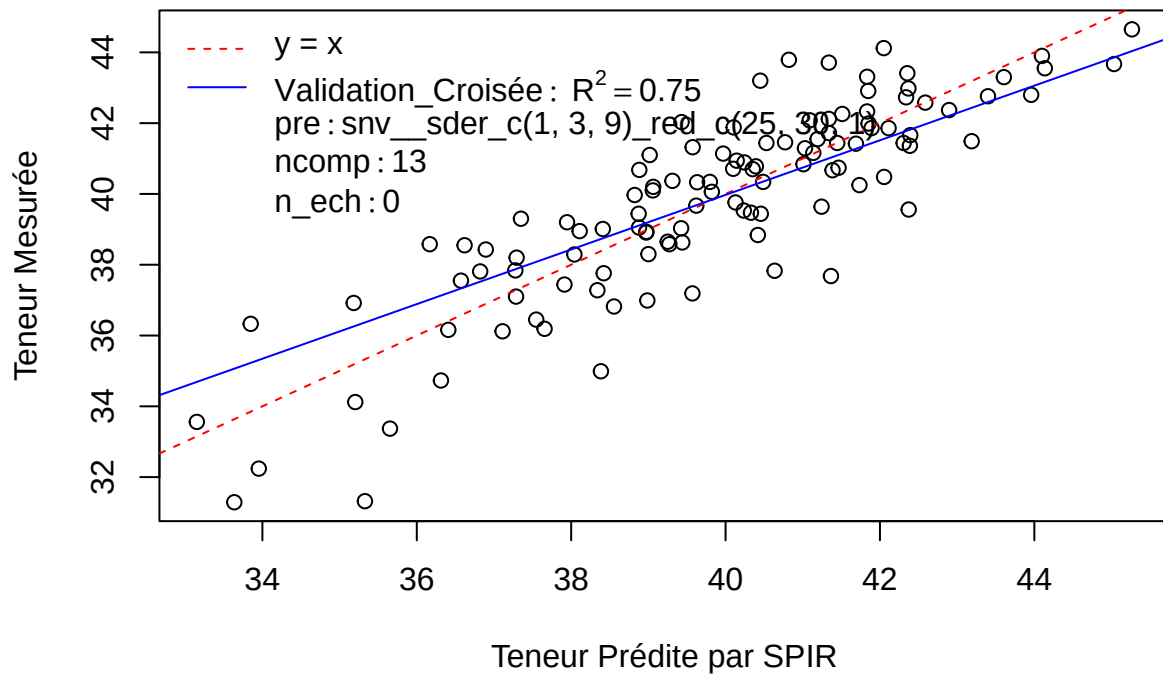
[1] "C16.0"

C16.0



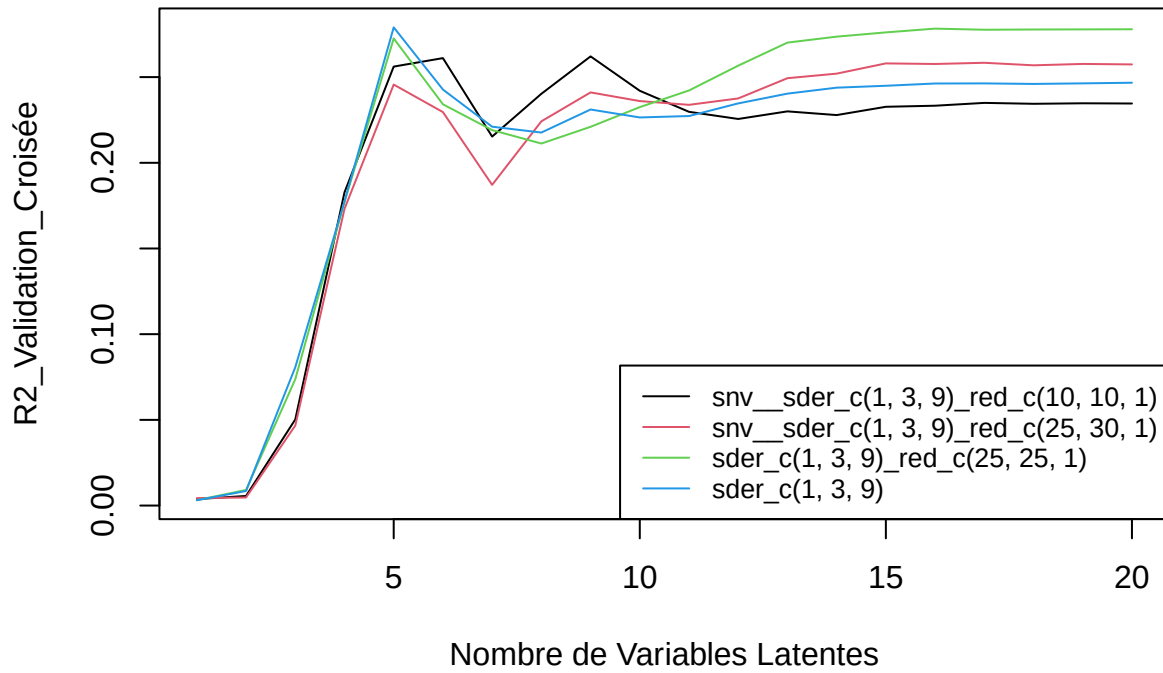
```
## [1] "snv__sder_c(1, 3, 9)_red_c(25, 30, 1)"
## [1] "R2"
## [1] 0.02 0.05 0.35 0.59 0.71 0.71 0.73 0.73 0.73 0.74 0.74 0.74 0.75 0.75 0.75 0.75 0.75 0.74 0.74
## [1] "SEP"
## y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1
## 2.73 2.77 2.22 1.76 1.47 1.48 1.43 1.43 1.42 1.40 1.39 1.39 1.38 1.38 1.38 1.38 1.38 1.39 1.39 1.39
```

C16.0



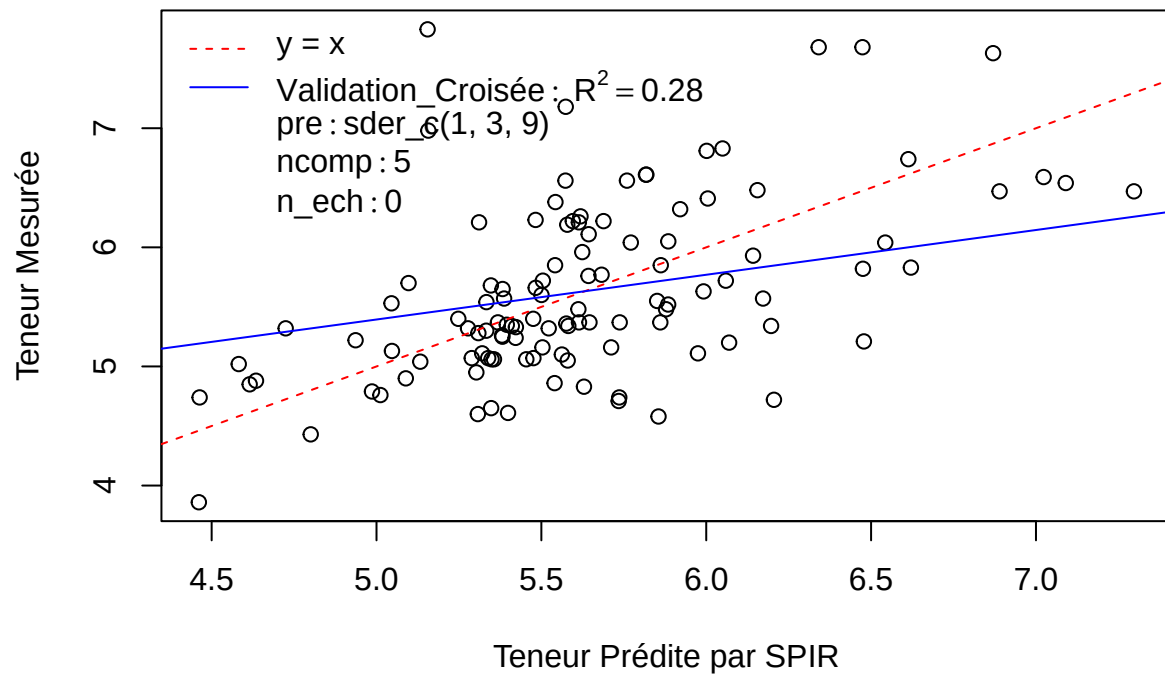
[1] "C18.0"

C18.0



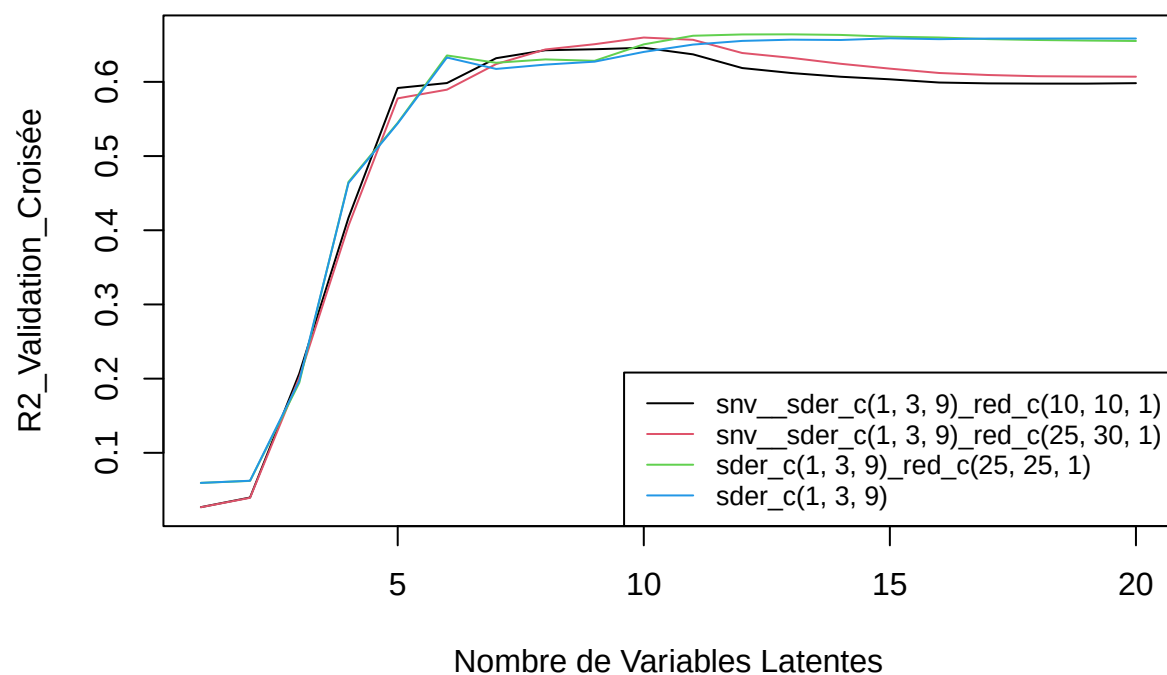
```
## [1] "sder_c(1, 3, 9)"
## [1] "R2"
## [1] 0.00 0.01 0.08 0.18 0.28 0.24 0.22 0.22 0.23 0.23 0.23 0.23 0.24 0.24 0.24 0.25 0.25 0.25 0.25
## [1] "SEP"
## y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1
## 0.75 0.82 0.72 0.69 0.64 0.67 0.68 0.68 0.68 0.68 0.69 0.69 0.68 0.68 0.68 0.68 0.68 0.68 0.68 0.68
```

C18.0



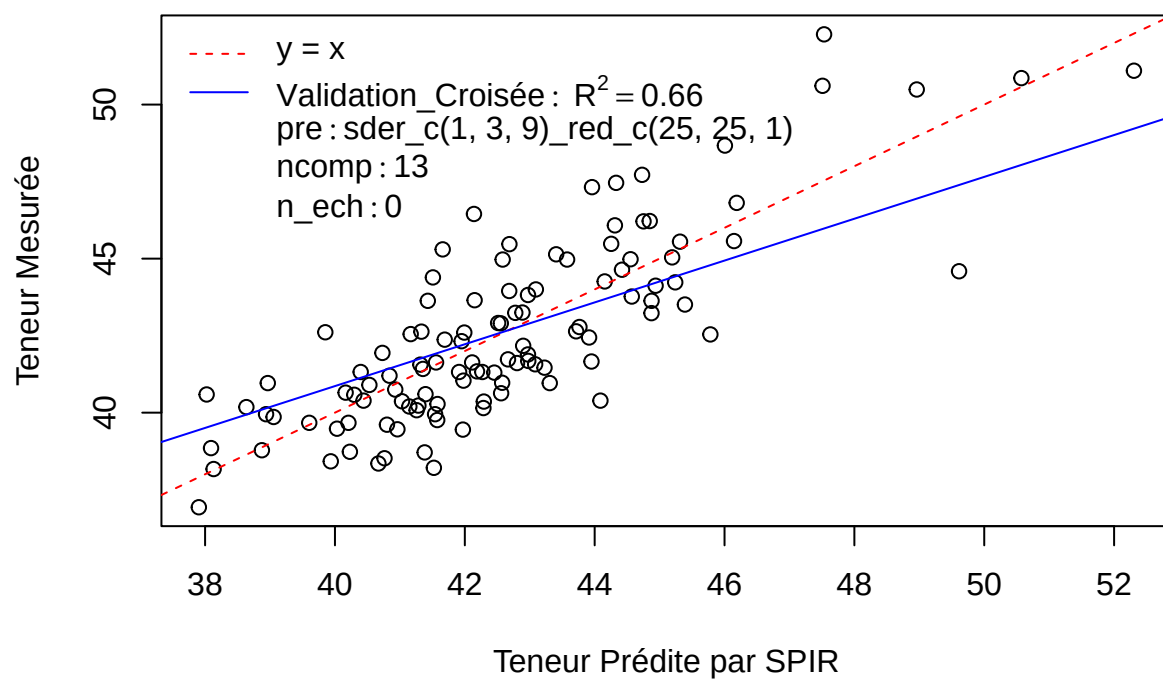
[1] "C18.1n9"

C18.1n9



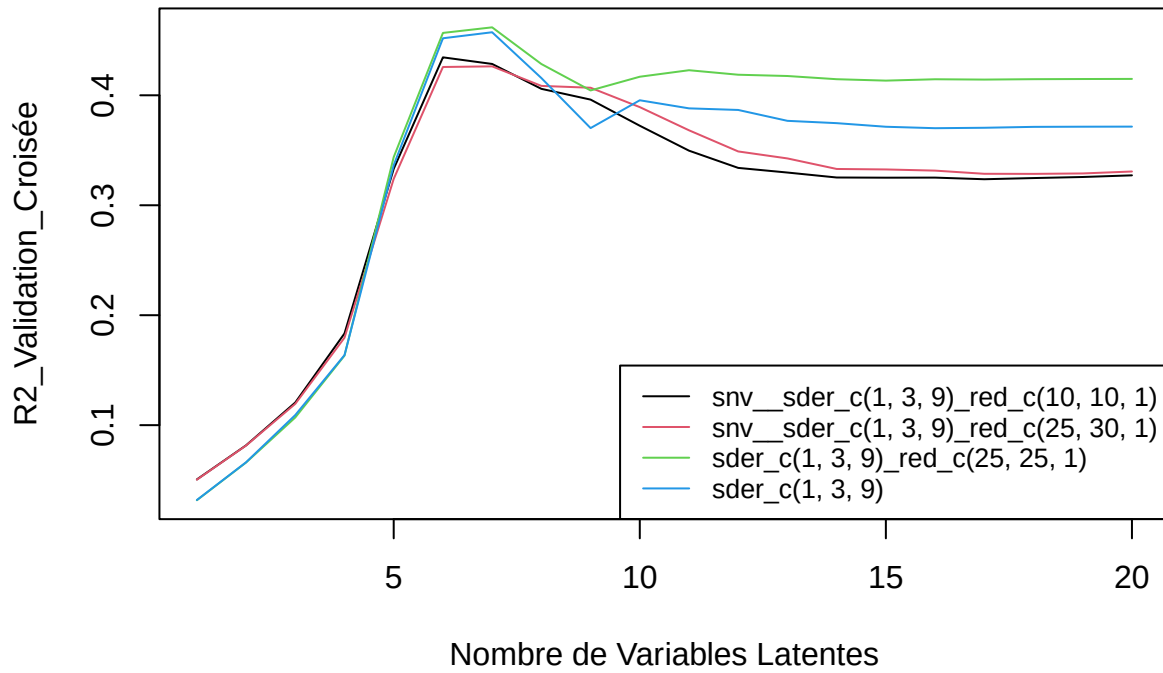
```
## [1] "sder_c(1, 3, 9)_red_c(25, 25, 1)"
## [1] "R2"
## [1] 0.06 0.06 0.19 0.47 0.54 0.64 0.63 0.63 0.63 0.65 0.66 0.66 0.66 0.66 0.66 0.66 0.66 0.66 0.66 0.66
## [1] "SEP"
## y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1
## 2.92 2.93 2.74 2.20 2.03 1.82 1.84 1.83 1.83 1.78 1.75 1.74 1.74 1.75 1.75 1.76 1.76 1.77 1.77 1.77
```


C18.1n9



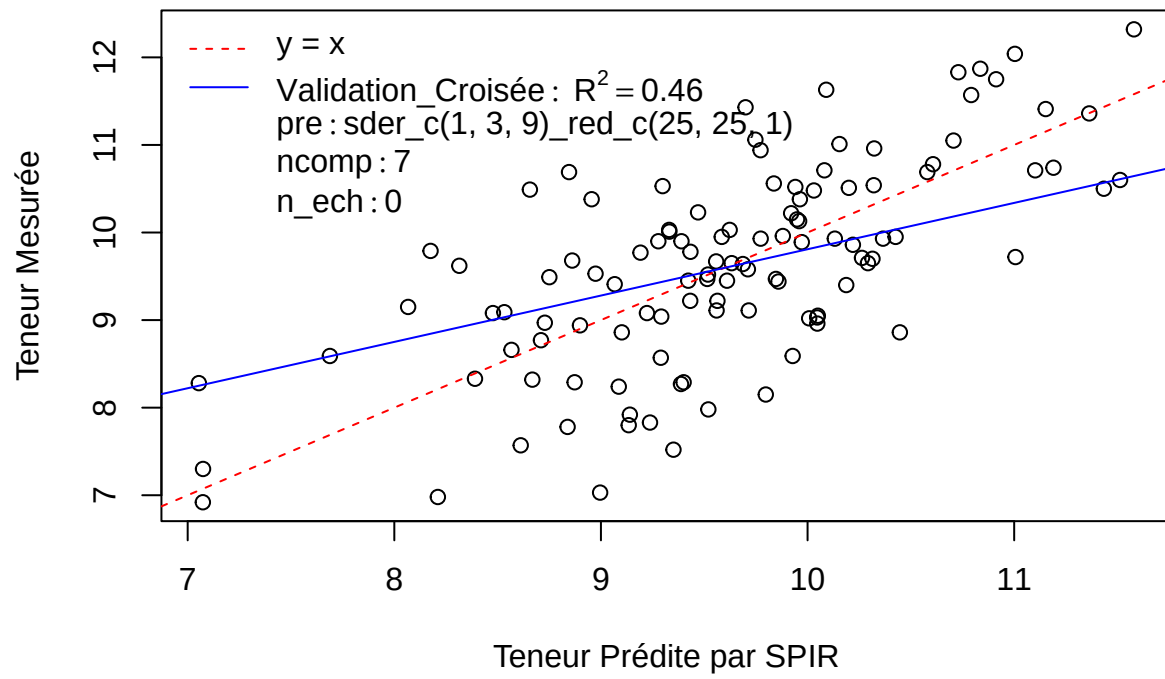
[1] "C18.2"

C18.2



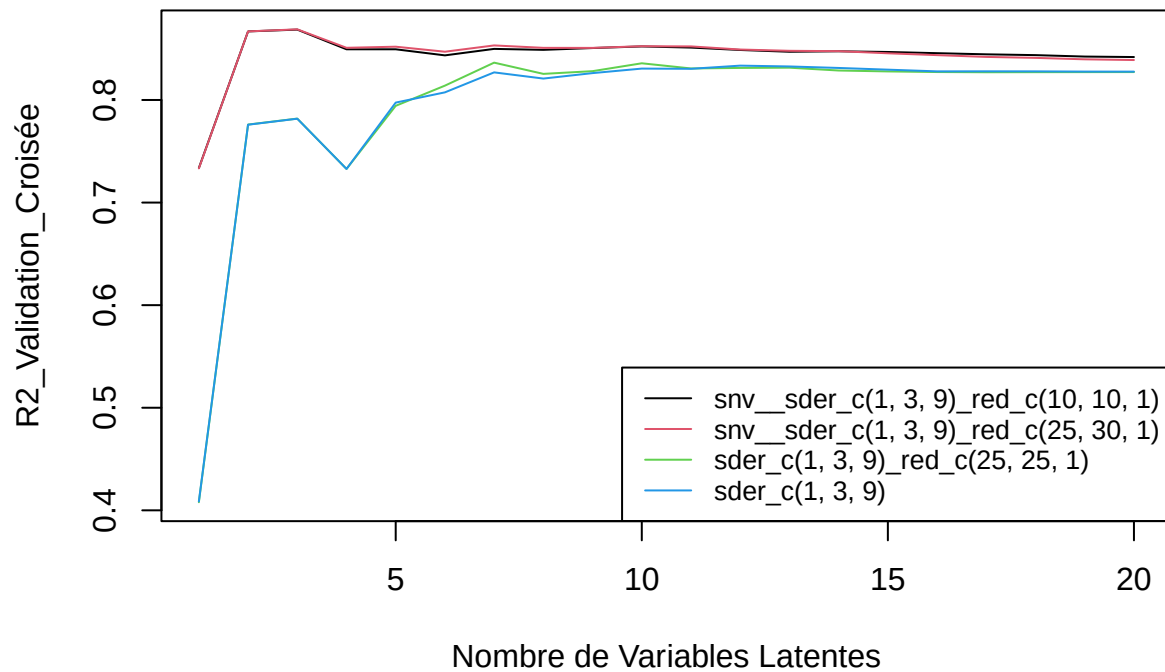
```
## [1] "sder_c(1, 3, 9)_red_c(25, 25, 1)"
## [1] "R2"
## [1] 0.03 0.07 0.11 0.16 0.34 0.46 0.46 0.43 0.40 0.42 0.42 0.42 0.42 0.41 0.41 0.41 0.41 0.41 0.41 0.41
## [1] "SEP"
## y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1
## 1.13 1.11 1.10 1.08 0.94 0.85 0.85 0.88 0.90 0.89 0.89 0.89 0.89 0.90 0.90 0.90 0.90 0.90 0.90 0.90
```

C18.2



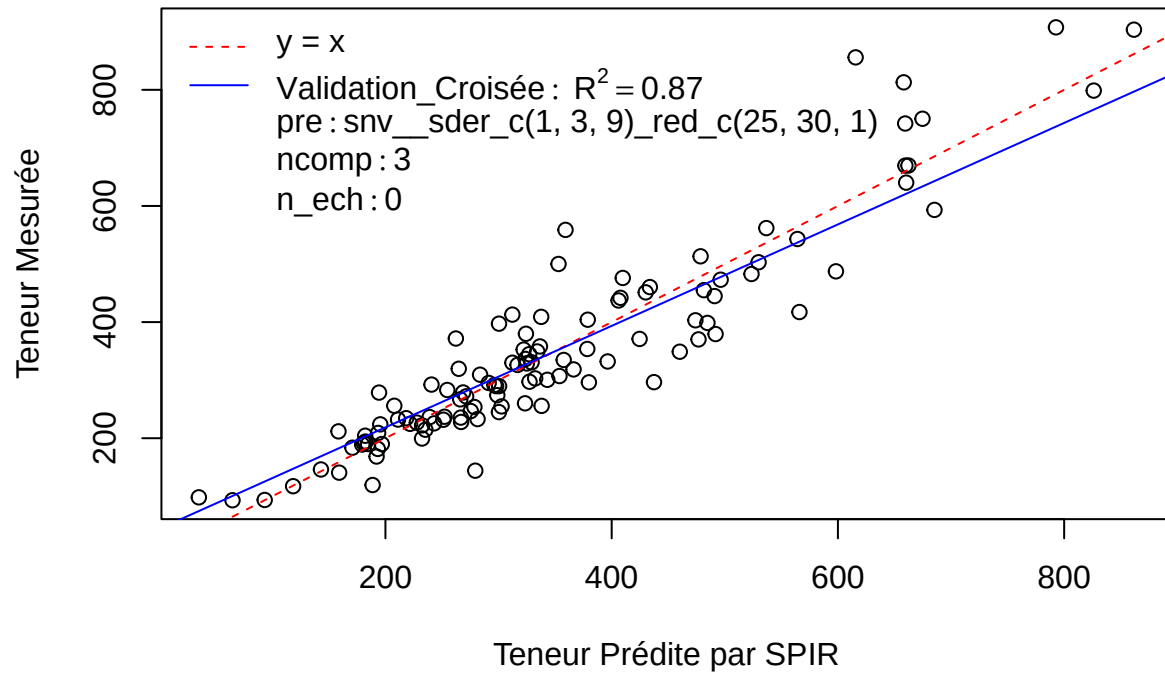
```
## [1] "trans.alpha.carotene"
```

trans.alpha.carotene



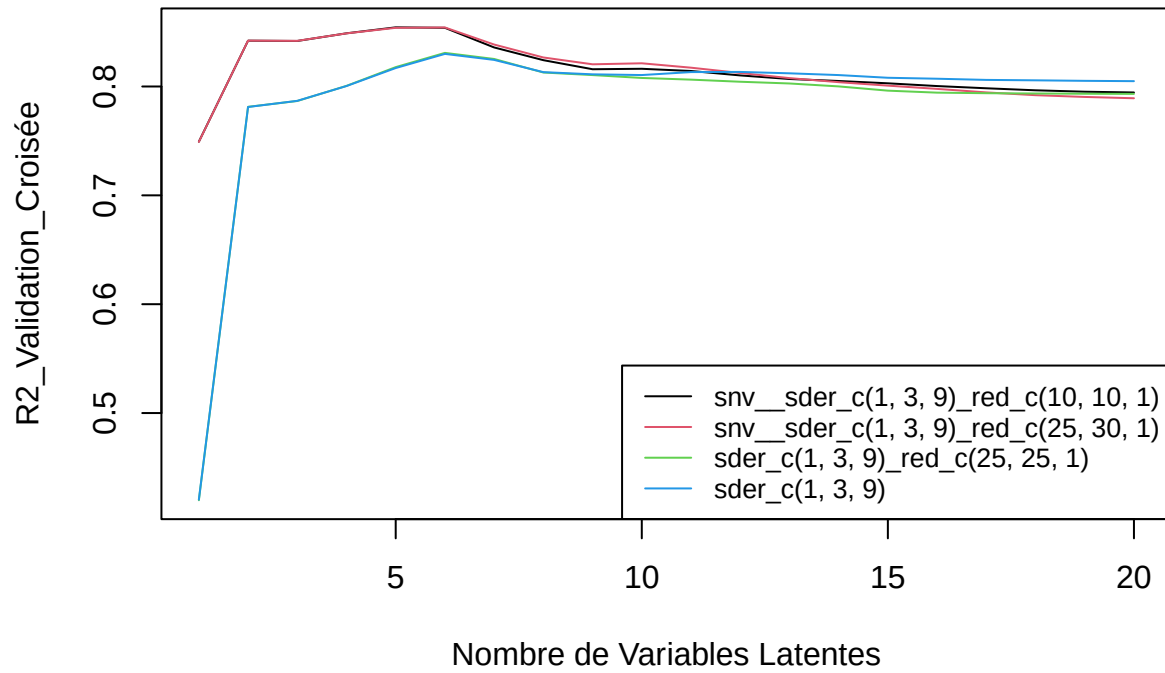
```
## [1] "snv__sder_c(1, 3, 9)_red_c(25, 30, 1)"
## [1] "R2"
## [1] 0.73 0.87 0.87 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.85 0.84 0.84 0.84 0.84
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 89.20 63.02 62.56 66.73 66.47 67.55 66.18 66.78 66.90 66.63 66.75 67.48 67.83 68.01 68.56 69.00 69.40
```

trans.alpha.carotene

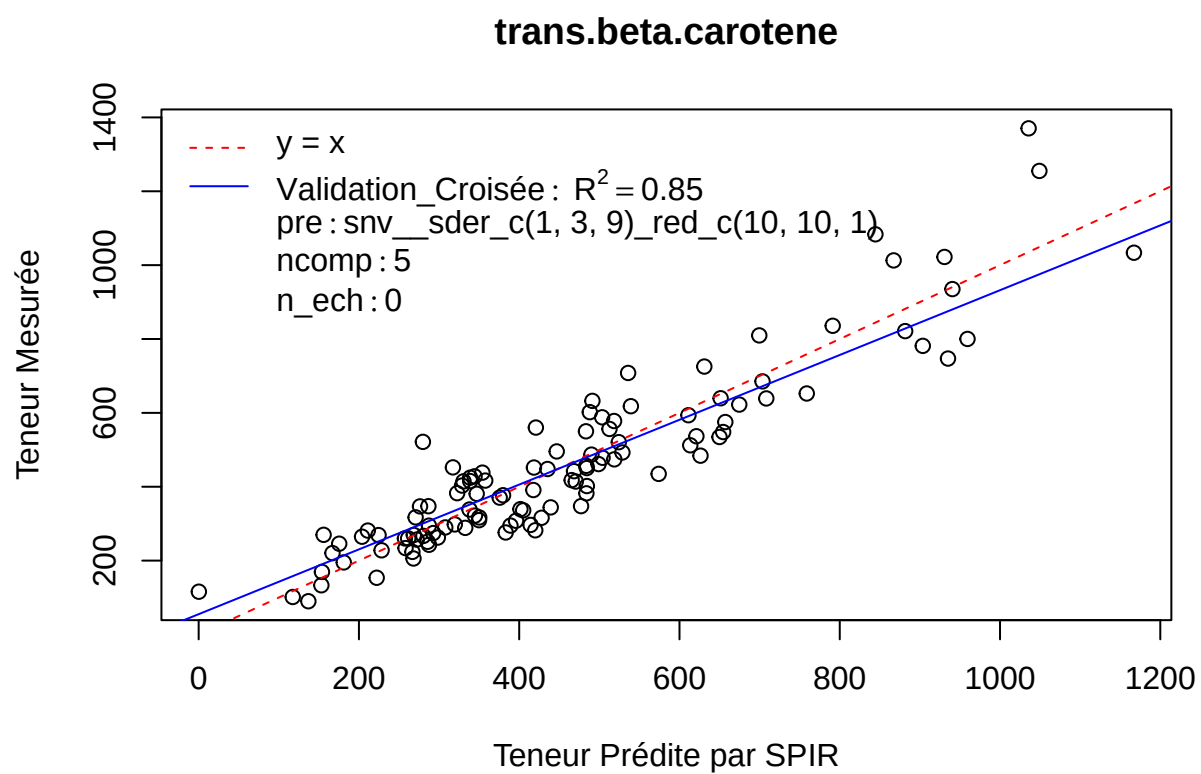


```
## [1] "trans.beta.carotene"
```

trans.beta.carotene

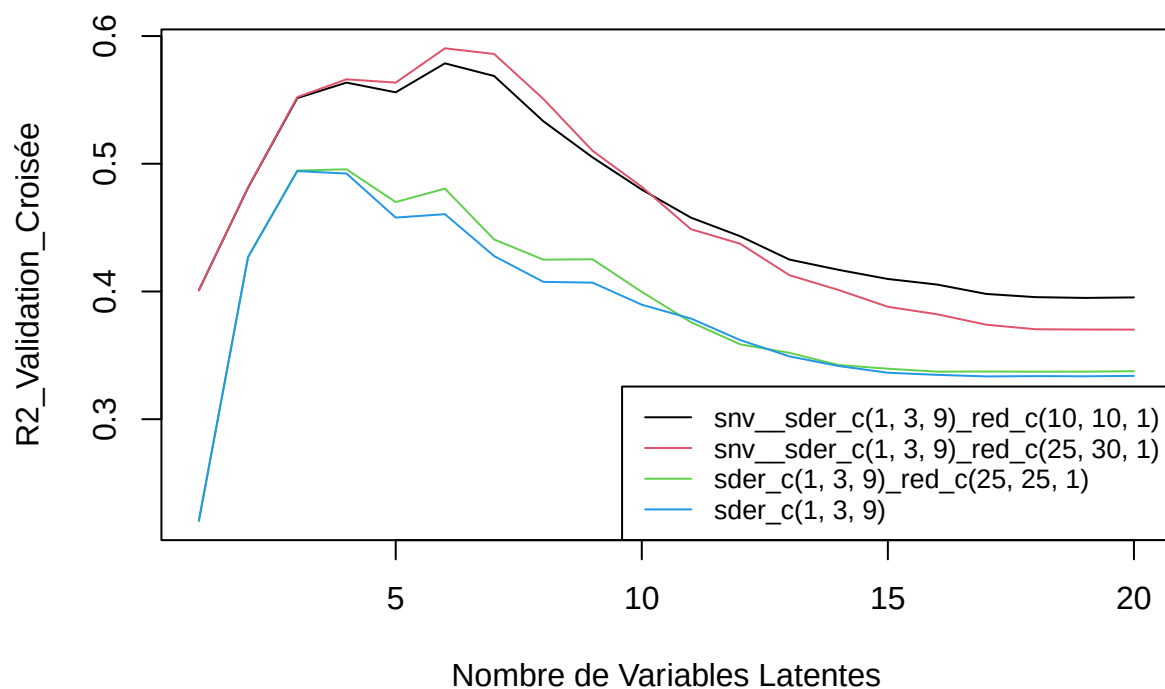


```
## [1] "snv__sder_c(1, 3, 9)_red_c(10, 10, 1)"
## [1] "R2"
## [1] 0.75 0.84 0.84 0.85 0.85 0.85 0.84 0.82 0.82 0.82 0.81 0.81 0.81 0.81 0.80 0.80 0.80 0.80 0.80
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 118.34  93.92  94.06  91.96  90.39  90.63  96.46 100.12 102.95 102.95 103.86 105.31 106.35 107.14 107.14
##      y1
## 110.62
```



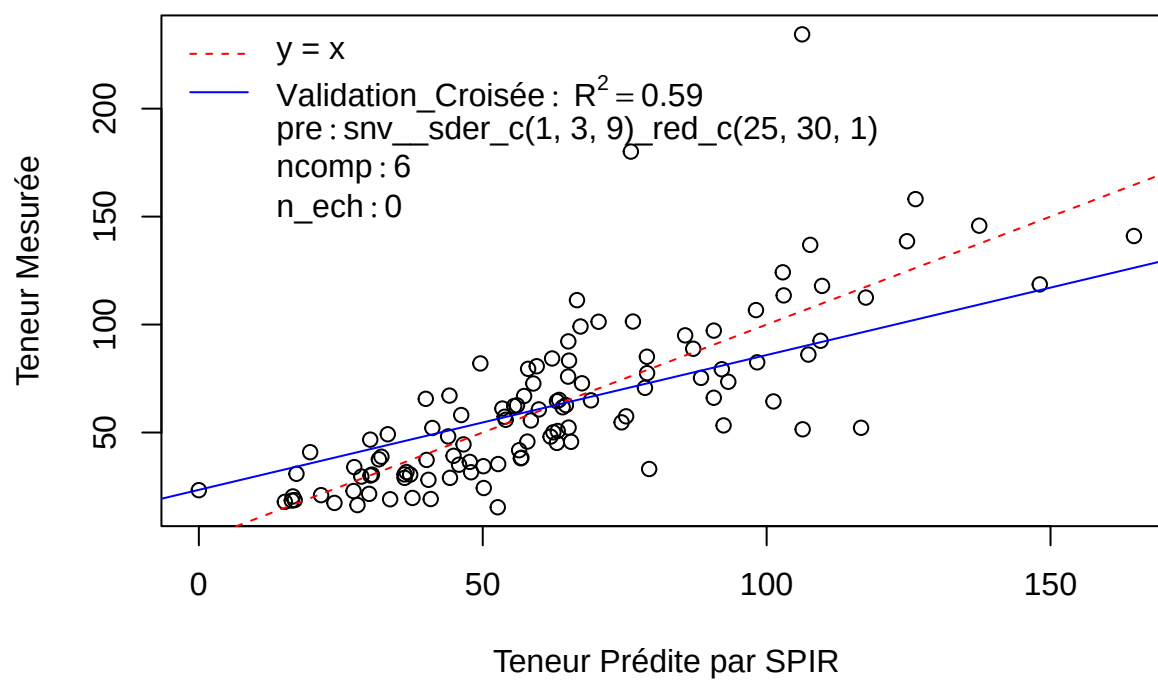
```
## [1] "X13.cis.beta.carotene"
```

X13.cis.beta.carotene



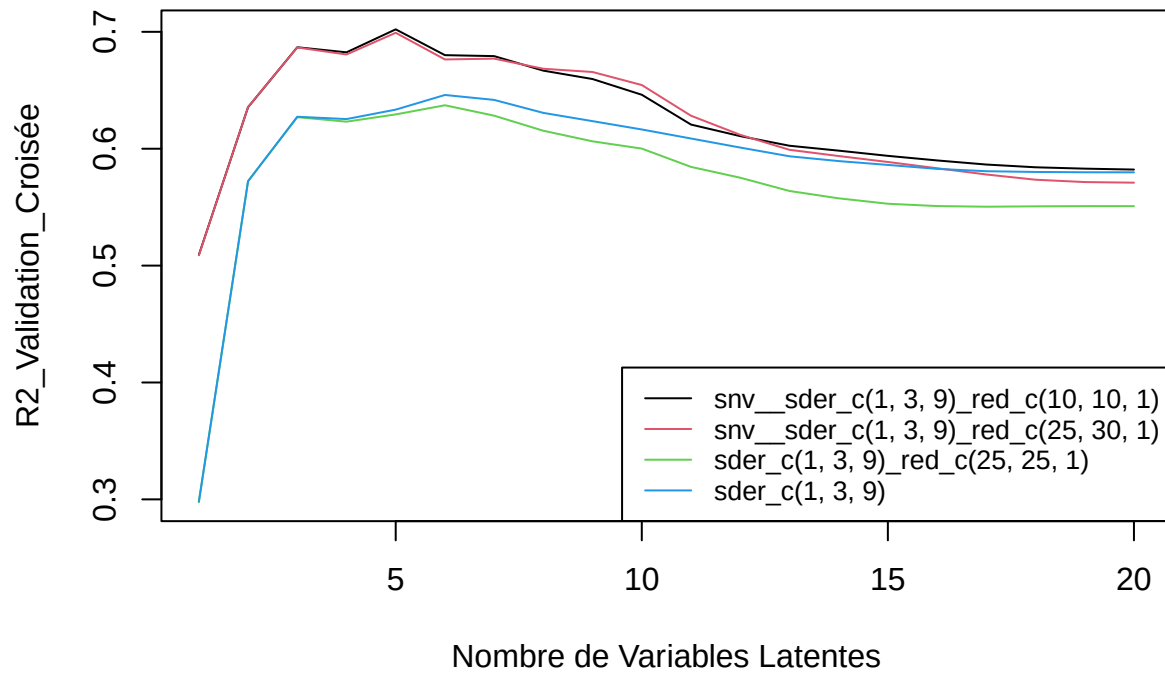
```
## [1] "snv__sder_c(1, 3, 9)_red_c(25, 30, 1)"
## [1] "R2"
## [1] 0.40 0.48 0.55 0.57 0.56 0.59 0.59 0.55 0.51 0.48 0.45 0.44 0.41 0.40 0.39 0.38 0.37 0.37 0.37 0.37
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 29.27 27.23 25.35 25.04 25.12 24.25 24.38 25.56 27.00 27.99 29.27 29.79 30.71 31.21 31.74 32.00 32.23
```


X13.cis.beta.carotene



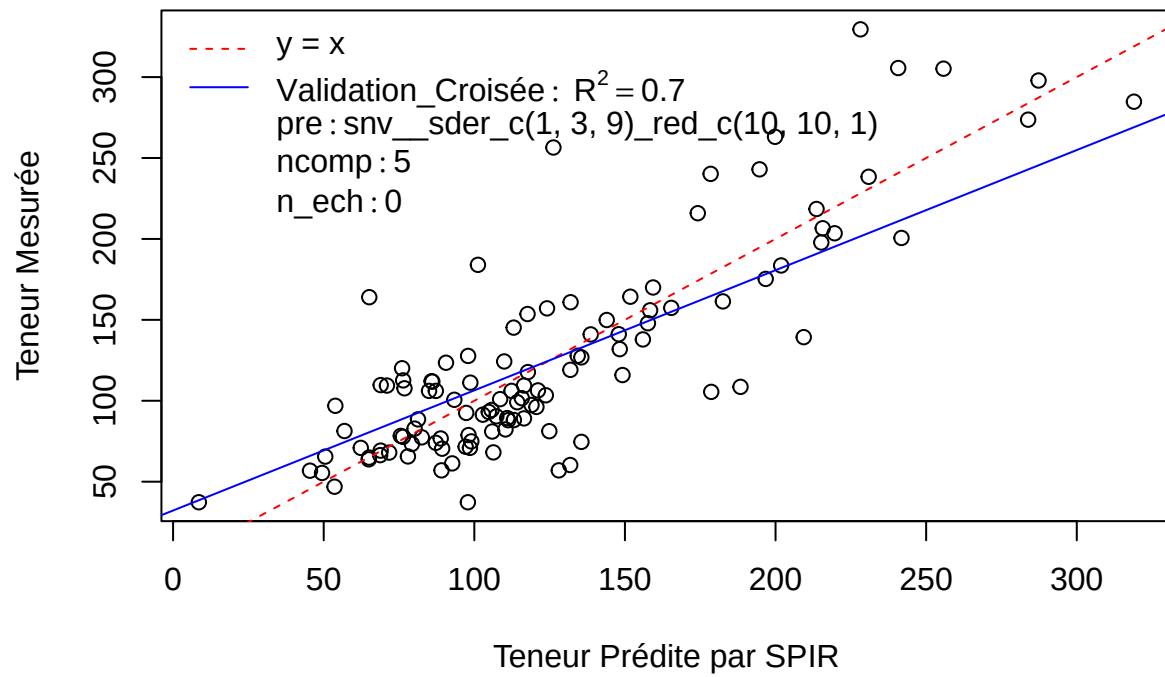
```
## [1] "X9.cis.beta.carotene"
```

X9.cis.beta.carotene

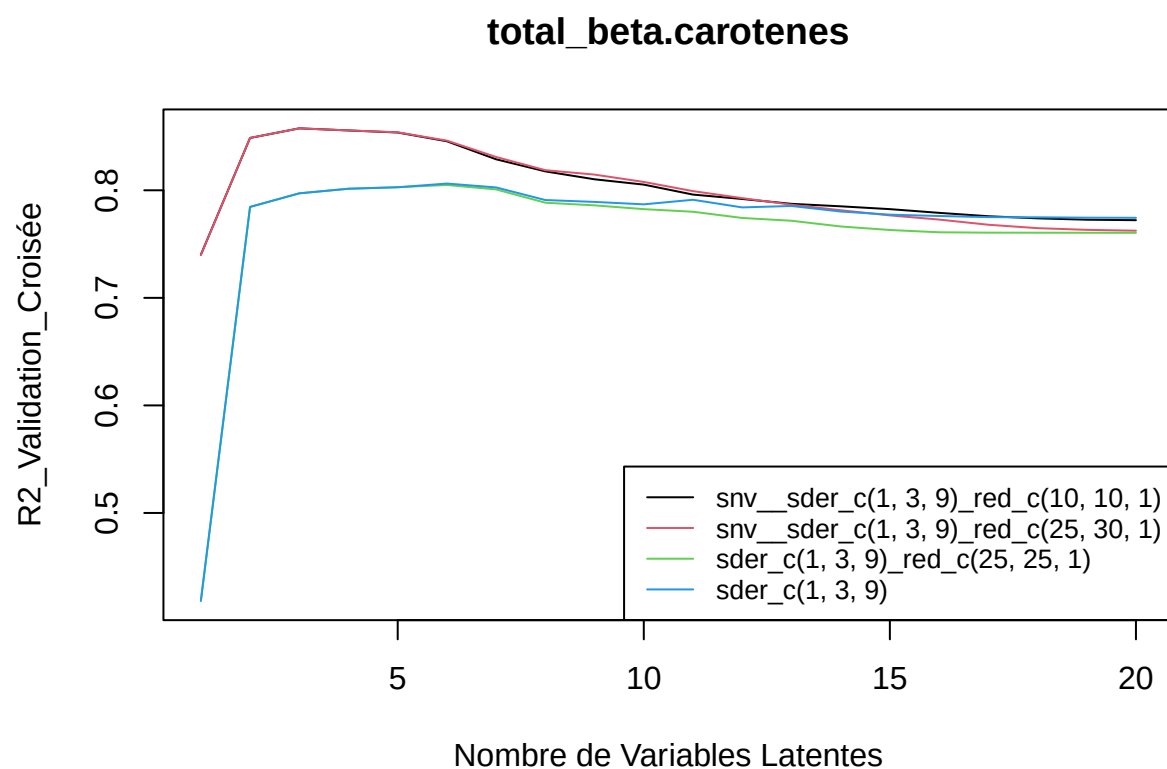


```
## [1] "snv__sder_c(1, 3, 9)_red_c(10, 10, 1)"
## [1] "R2"
## [1] 0.51 0.64 0.69 0.68 0.70 0.68 0.68 0.67 0.66 0.65 0.62 0.61 0.60 0.60 0.59 0.59 0.59 0.58 0.58 (
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 44.94 38.73 35.91 36.30 35.14 36.54 36.63 37.42 38.04 39.03 40.69 41.47 42.04 42.37 42.68 42.97 43.2
```

X9.cis.beta.carotene

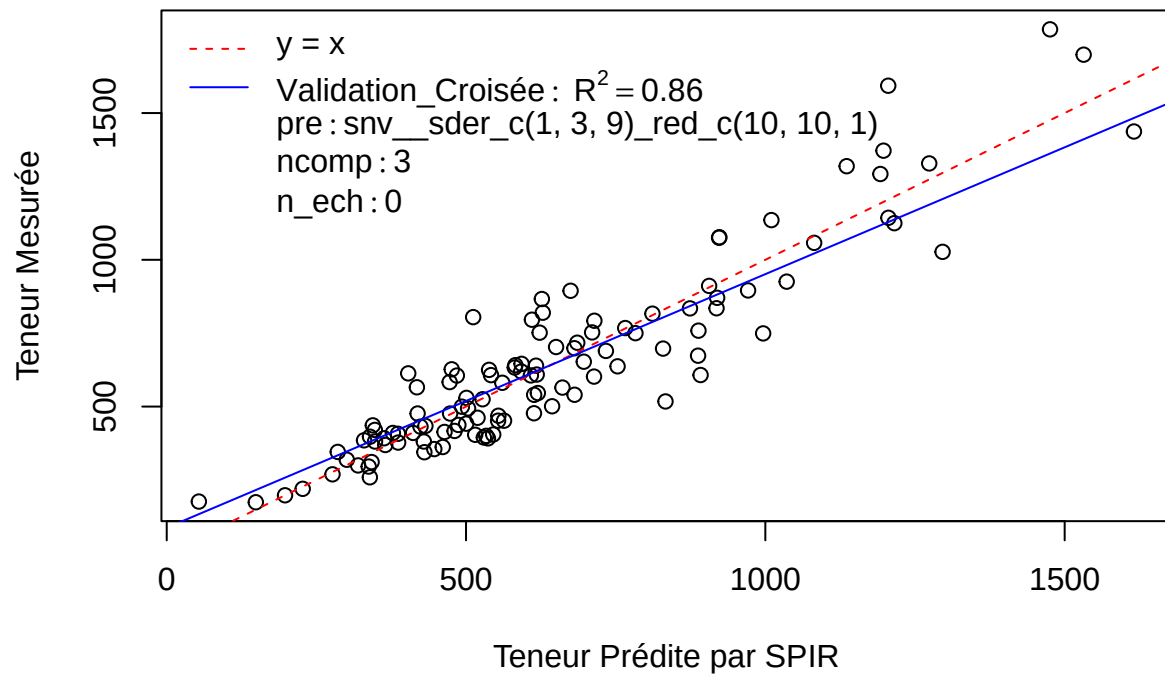


```
## [1] "total_beta.carotenes"
```

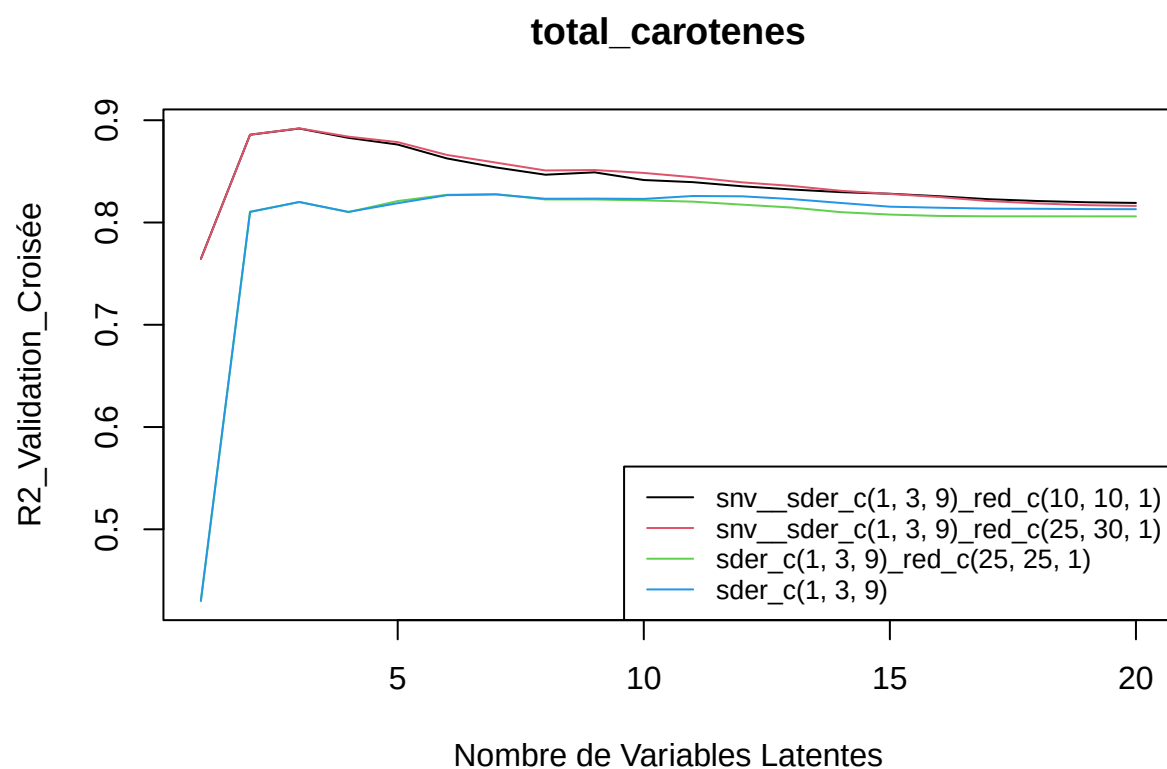


```
## [1] "snv__sder_c(1, 3, 9)_red_c(10, 10, 1)"
## [1] "R2"
## [1] 0.74 0.85 0.86 0.86 0.85 0.85 0.83 0.82 0.81 0.81 0.80 0.79 0.79 0.79 0.78 0.78 0.78 0.77 0.77
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 162.75 124.19 120.48 121.42 122.31 125.86 132.87 137.56 141.04 143.05 146.96 148.94 150.78 151.97 153.40
##      y1
## 157.40
```

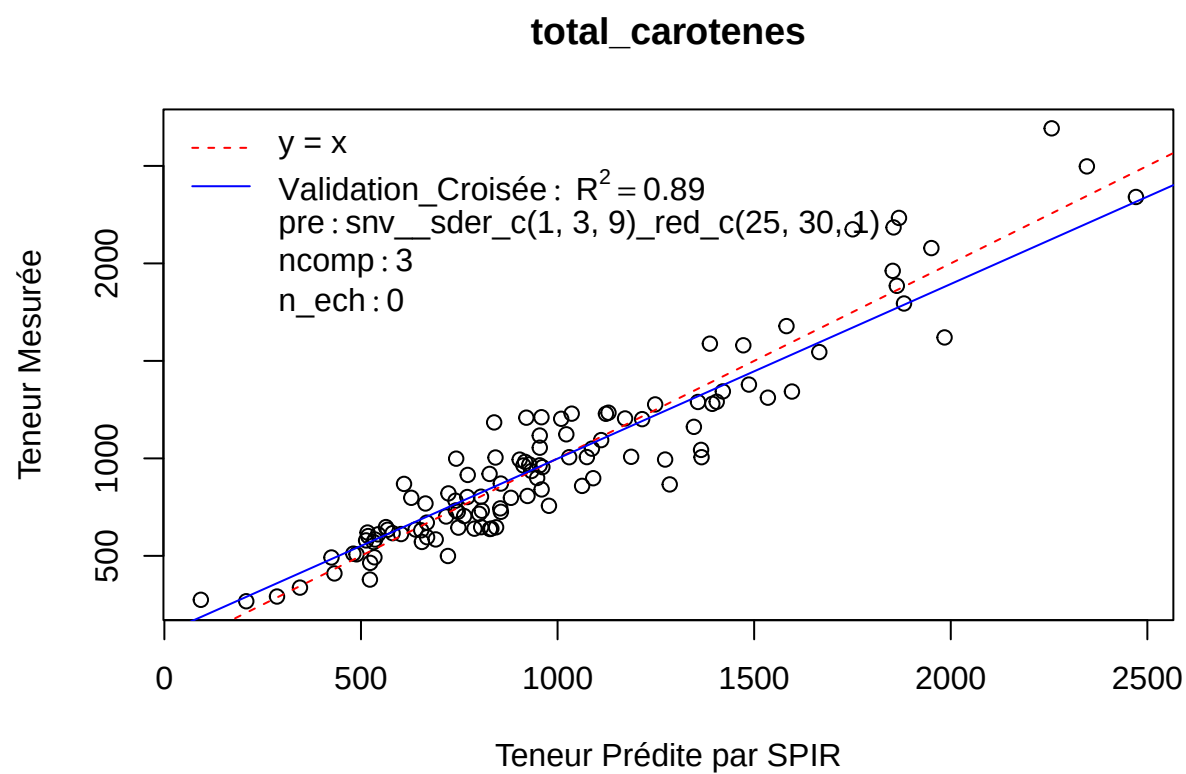
total_beta.carotenes



```
## [1] "total_carotenes"
```

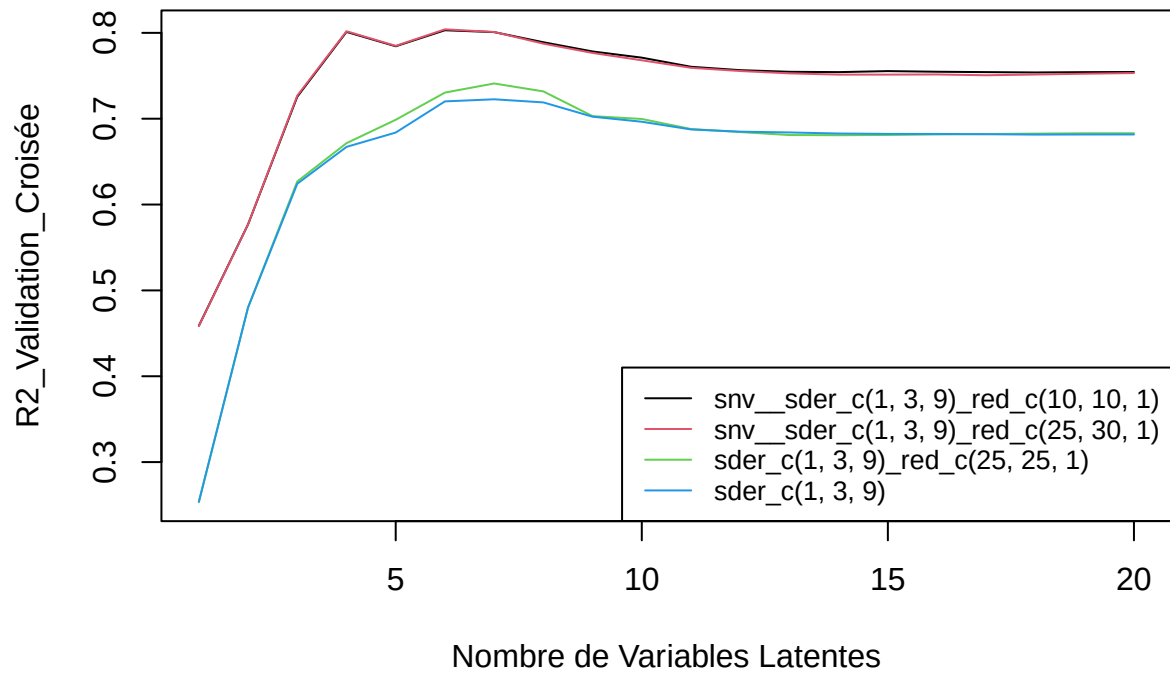


```
## [1] "snv__sder_c(1, 3, 9)_red_c(25, 30, 1)"
## [1] "R2"
## [1] 0.76 0.89 0.89 0.88 0.88 0.87 0.86 0.85 0.85 0.85 0.84 0.84 0.84 0.83 0.83 0.82 0.82 0.82 0.82
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 234.92 163.40 158.92 164.87 168.71 177.29 182.34 187.97 187.93 190.22 193.05 196.74 199.10 202.62 204.92
##      y1
## 212.52
```



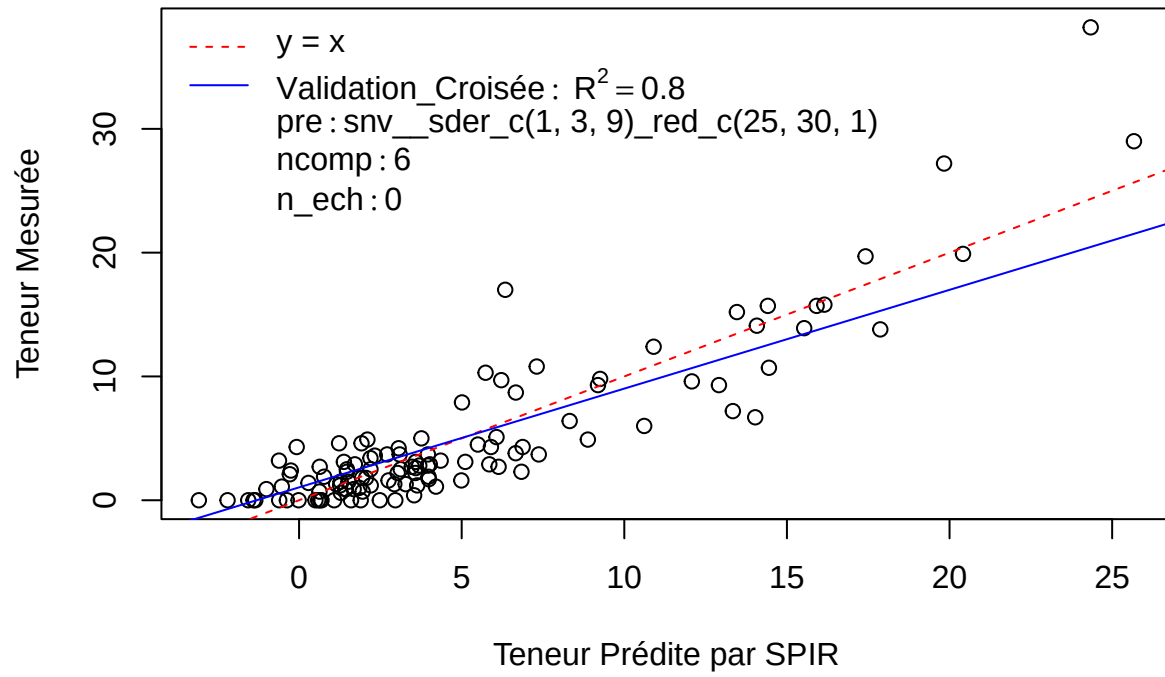
```
## [1] "Lycopene"
```

Lycopene

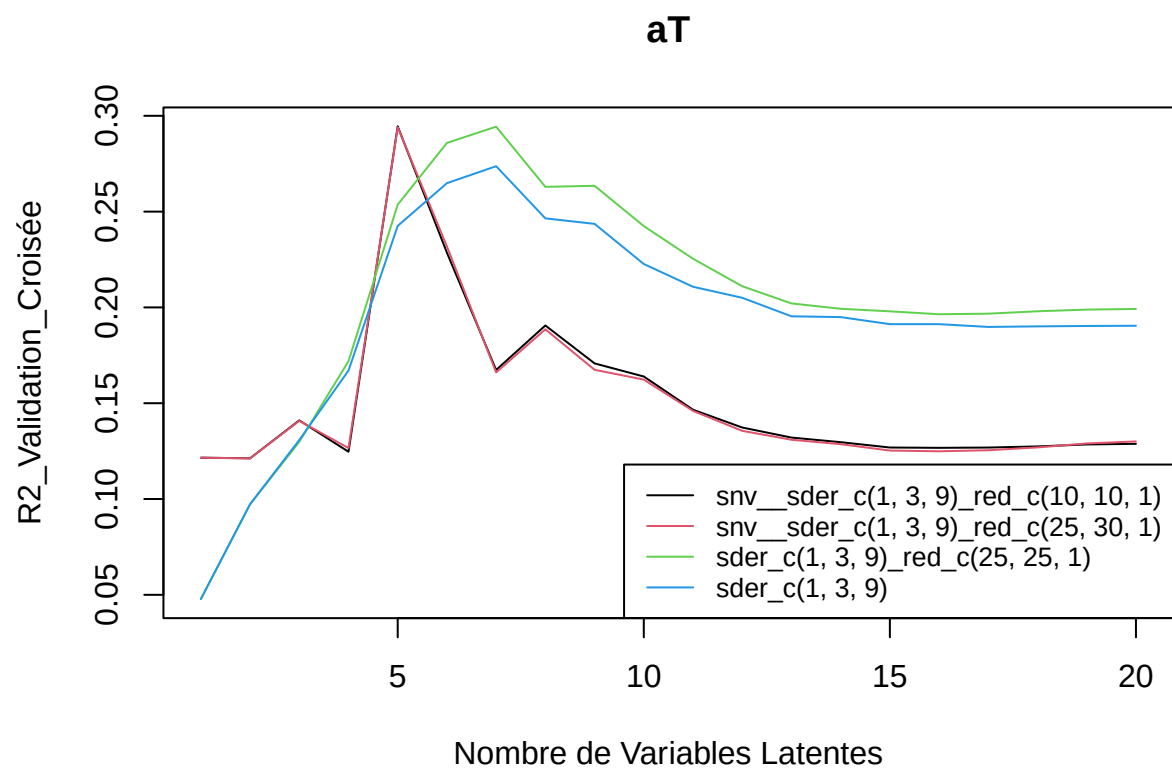


```
## [1] "snv_sder_c(1, 3, 9)_red_c(25, 30, 1)"
## [1] "R2"
## [1] 0.46 0.58 0.73 0.80 0.78 0.80 0.80 0.80 0.79 0.78 0.77 0.76 0.76 0.75 0.75 0.75 0.75 0.75 0.75 0.75
## [1] "SEP"
##   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1   y1
## 4.74 4.19 3.37 2.87 2.99 2.85 2.87 2.97 3.05 3.11 3.17 3.20 3.22 3.23 3.23 3.23 3.23 3.23 3.22 3.22
```


Lycopene

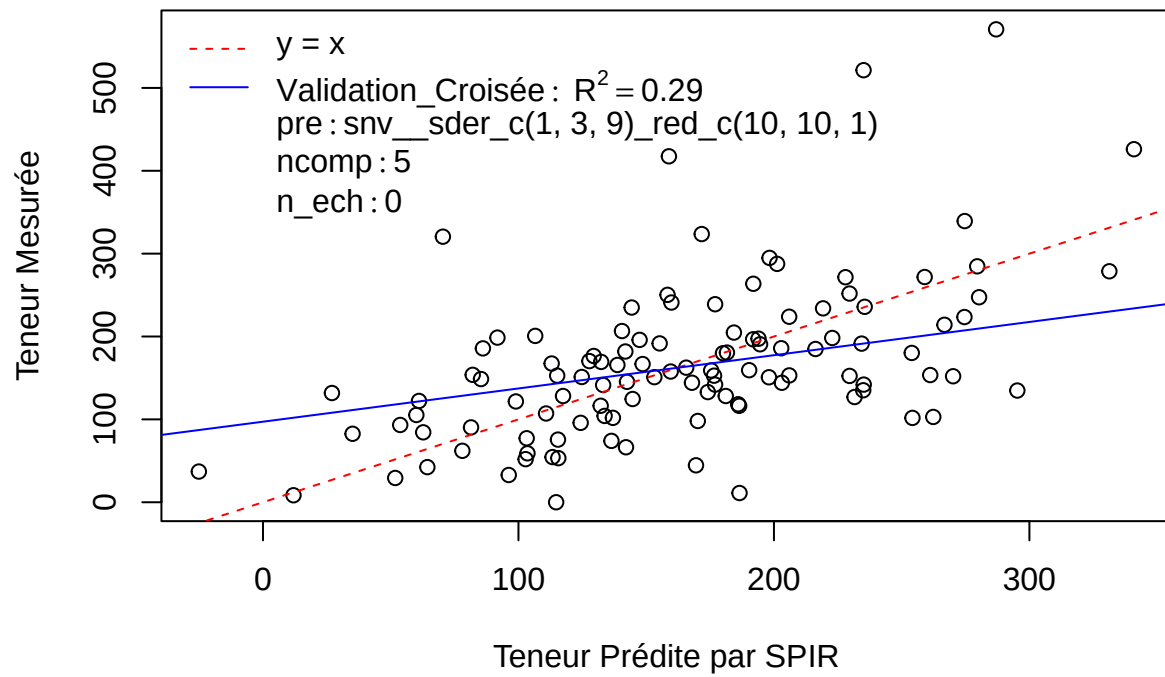


```
## [1] "aT"
```

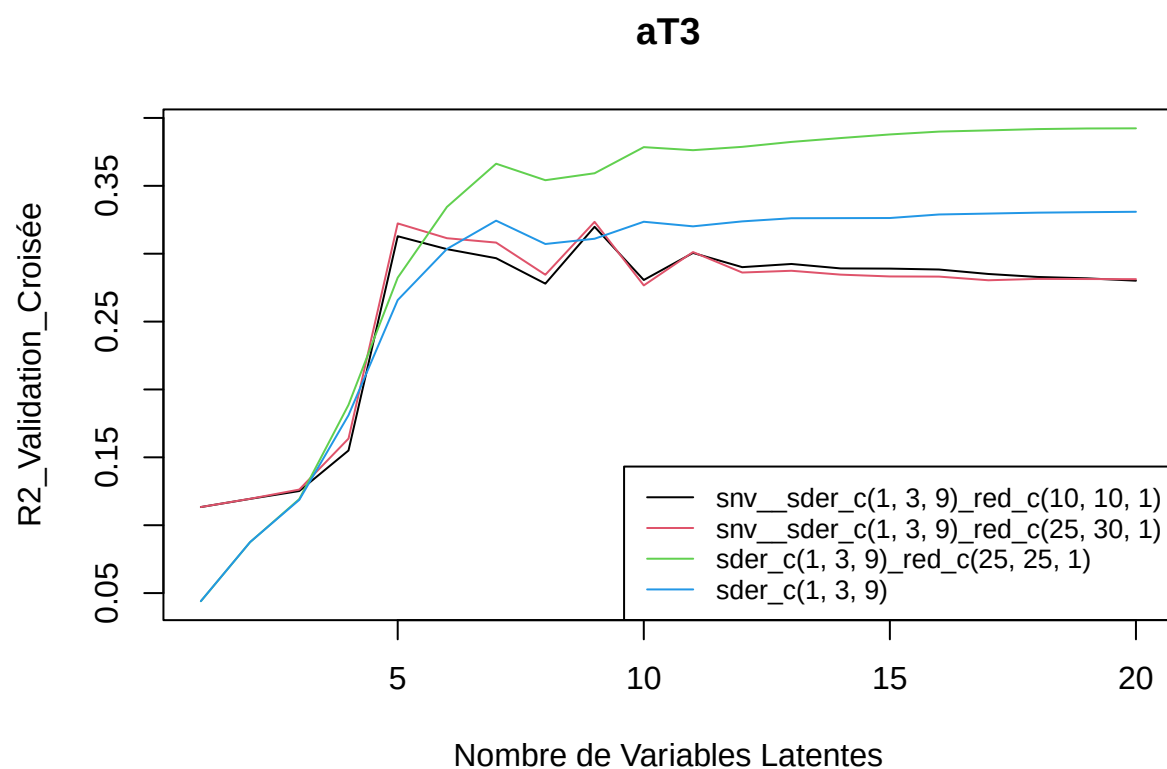


```
## [1] "snv__sder_c(1, 3, 9)_red_c(10, 10, 1)"
## [1] "R2"
## [1] 0.12 0.12 0.14 0.12 0.29 0.23 0.17 0.19 0.17 0.16 0.15 0.14 0.13 0.13 0.13 0.13 0.13 0.13 0.13 0.13
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 88.61 88.75 88.27 92.98 81.38 86.02 91.67 91.48 94.52 96.78 99.95 102.38 104.02 104.88 105.52
##      y1
## 105.52
```

aT

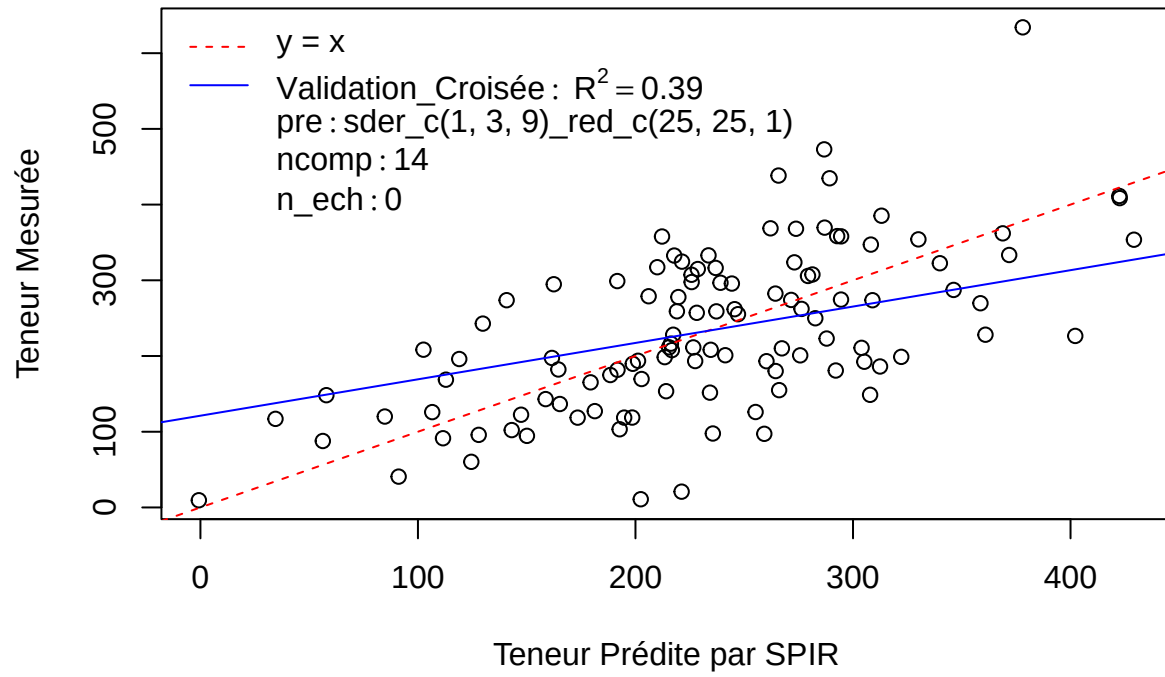


```
## [1] "aT3"
```

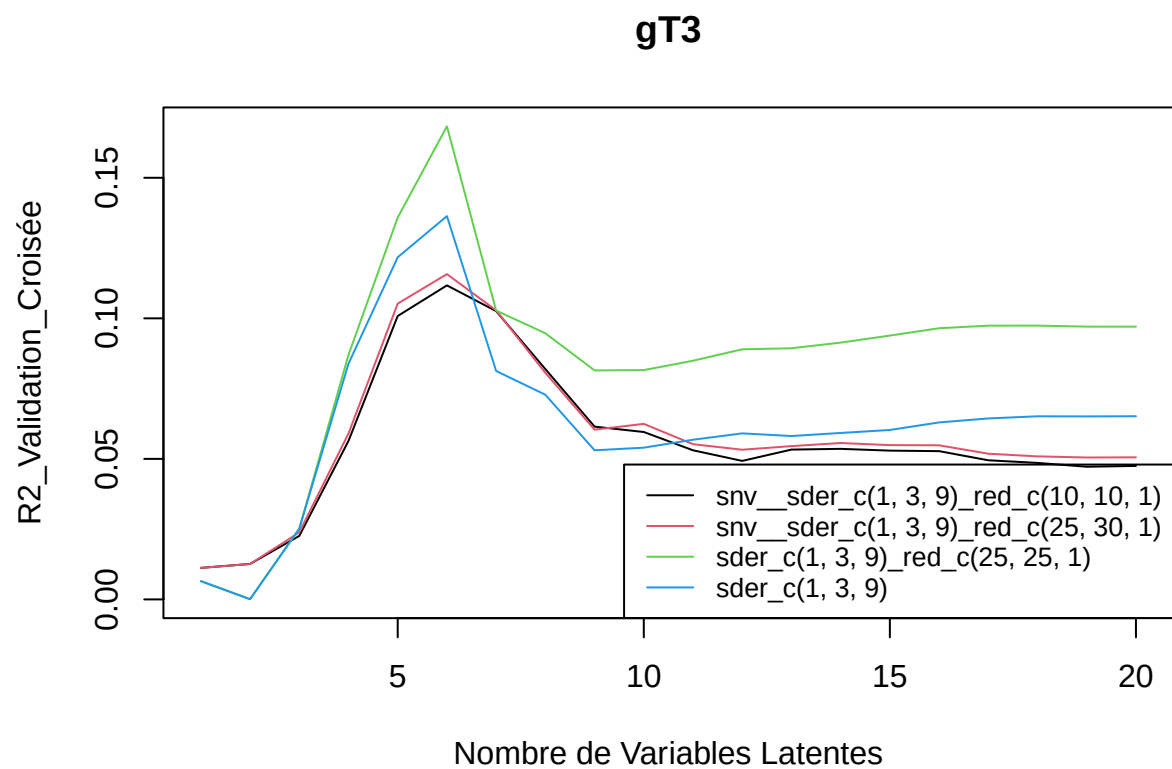


```
## [1] "sder_c(1, 3, 9)_red_c(25, 25, 1)"
## [1] "R2"
## [1] 0.04 0.09 0.12 0.19 0.28 0.33 0.37 0.35 0.36 0.38 0.38 0.38 0.38 0.39 0.39 0.39 0.39 0.39 0.39
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 103.80 101.47 100.66  97.32  91.27  88.10  85.17  86.19  86.01  84.80  85.19  85.08  84.77  84.56  84.35
##      y1
##  84.00
```

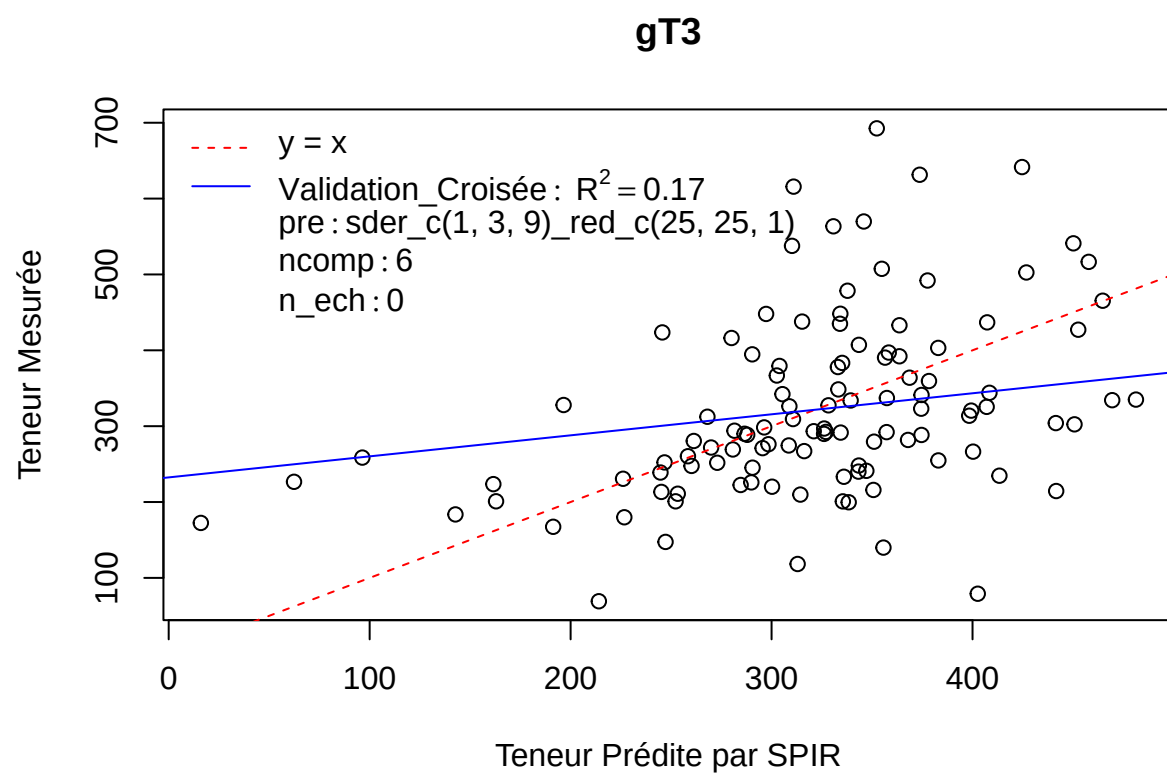
aT3



[1] "gT3"

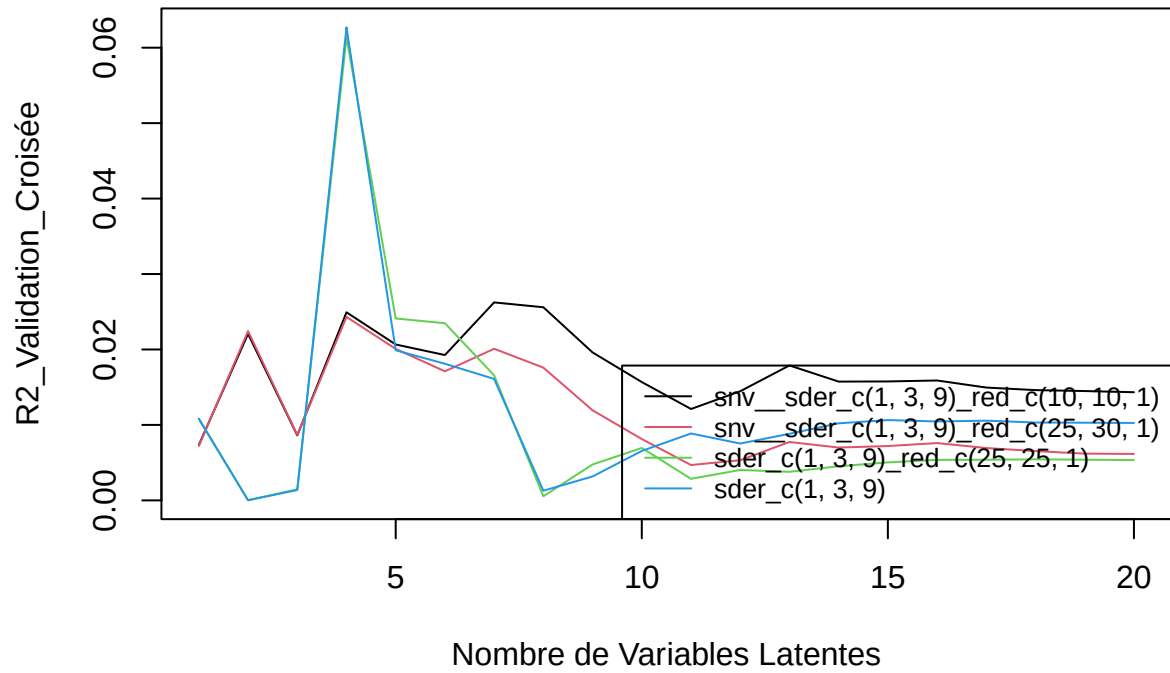


```
## [1] "sder_c(1, 3, 9)_red_c(25, 25, 1)"
## [1] "R2"
## [1] 0.01 0.00 0.03 0.09 0.14 0.17 0.10 0.09 0.08 0.08 0.08 0.09 0.09 0.09 0.09 0.10 0.10 0.10 0.10 0.10
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 120.36 120.24 121.57 116.34 113.70 112.76 123.00 124.27 126.67 127.31 127.81 127.34 127.18 126.85 126.04 126.04
##      y1
## 126.04
```



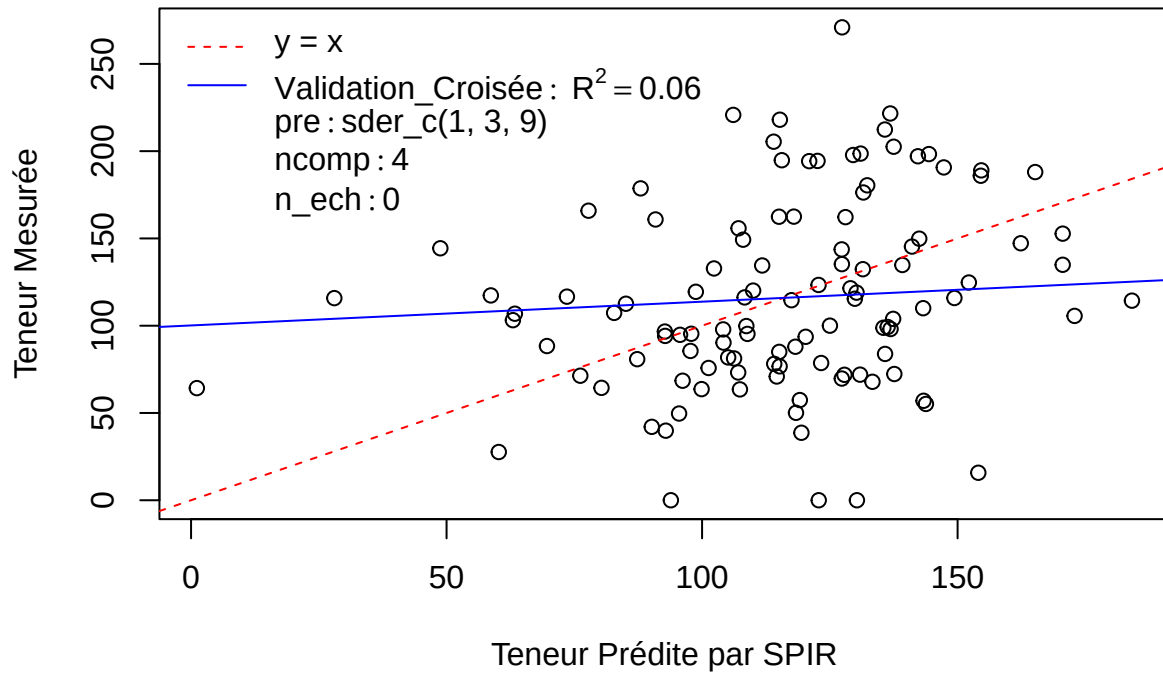
```
## [1] "dT3"
```

dT3

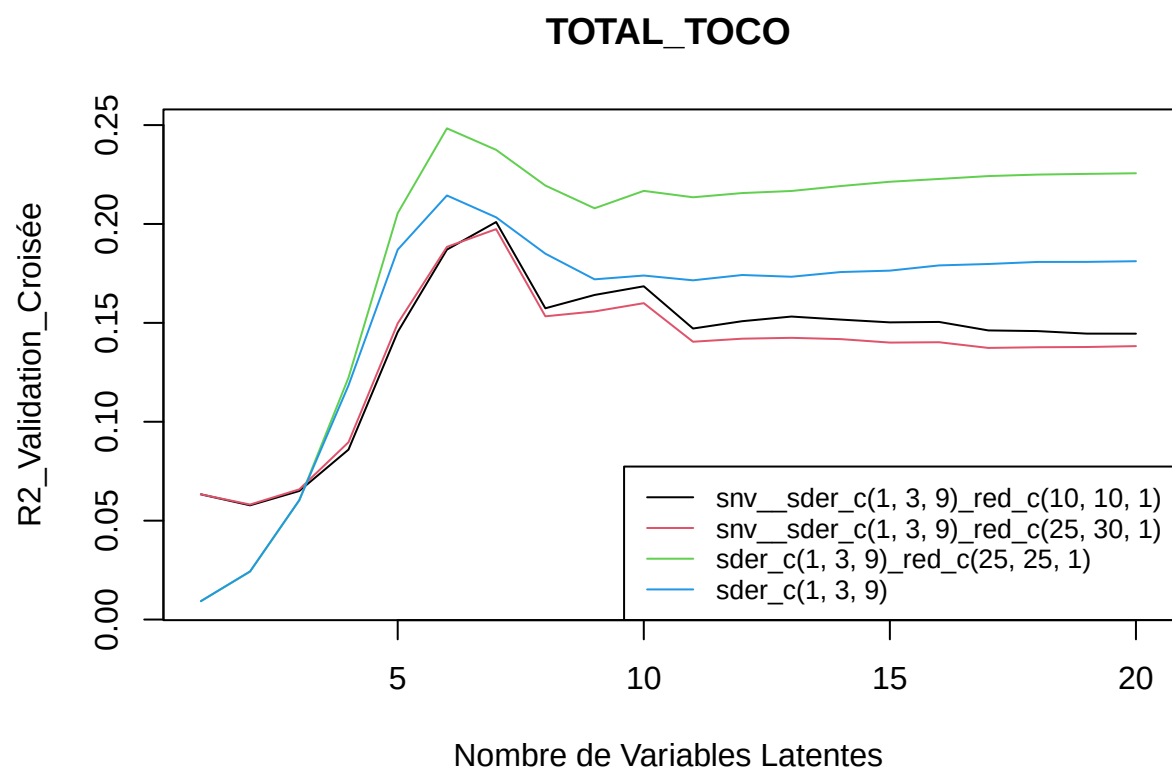


```
## [1] "sder_c(1, 3, 9)"
## [1] "R2"
## [1] 0.01 0.00 0.00 0.06 0.02 0.02 0.02 0.00 0.00 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01 0.01
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 55.13 54.57 57.00 54.49 58.93 60.76 62.31 65.50 65.13 64.68 64.75 64.80 64.73 64.53 64.48 64.57 64.53
```

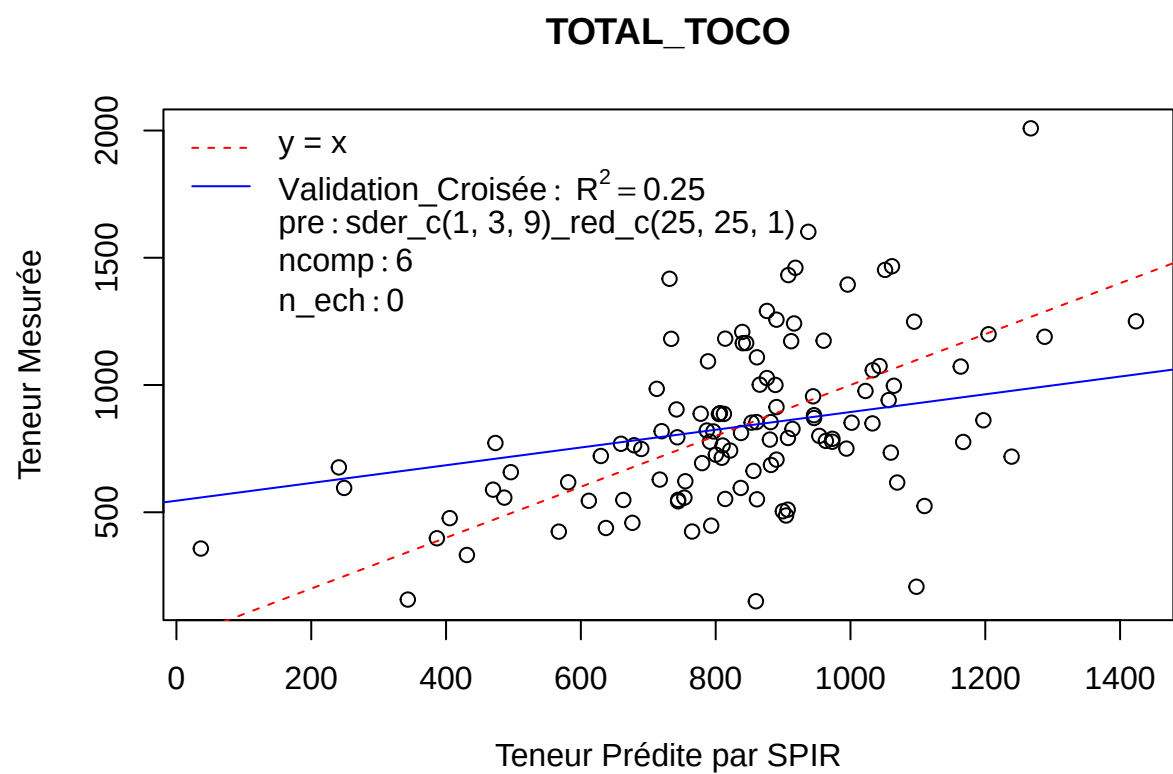

dT3



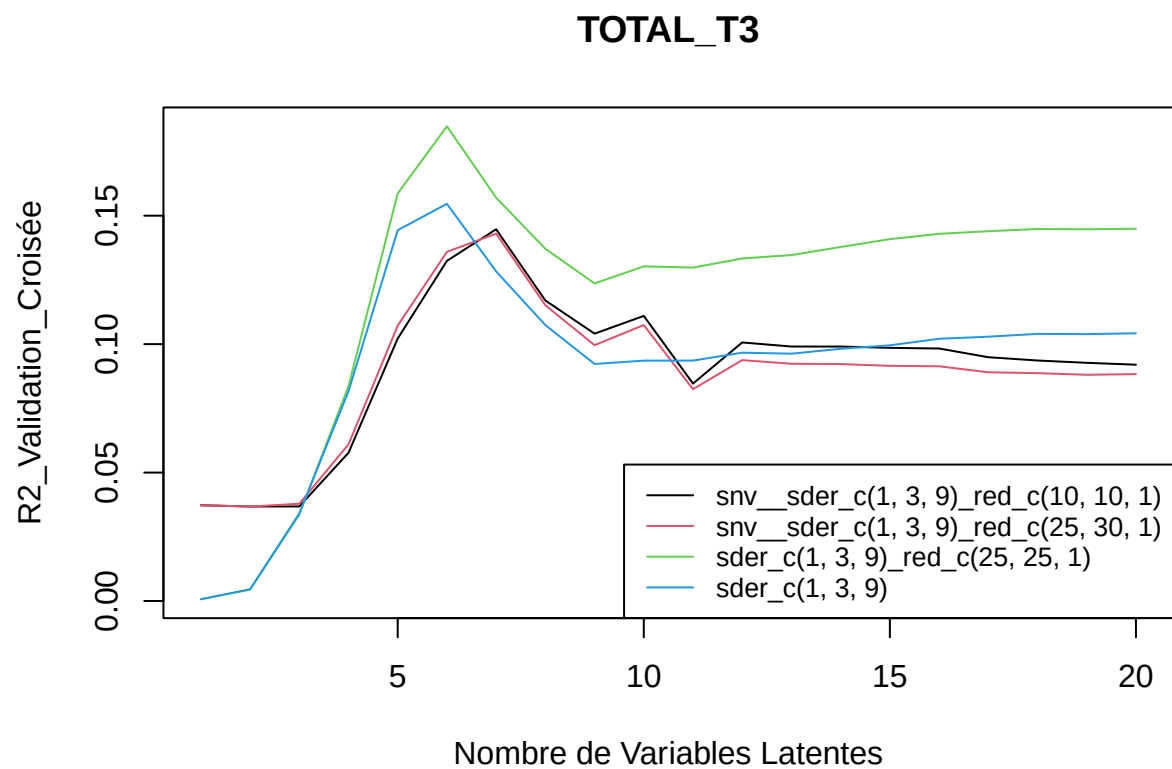
```
## [1] "TOTAL_TOCO"
```



```
## [1] "sder_c(1, 3, 9)_red_c(25, 25, 1)"
## [1] "R2"
## [1] 0.01 0.02 0.06 0.12 0.21 0.25 0.24 0.22 0.21 0.22 0.21 0.22 0.22 0.22 0.22 0.22 0.22 0.23
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 317.61 315.51 314.13 304.33 288.53 282.28 286.63 291.82 296.60 295.57 298.06 298.01 297.63 297.26 297.26
##      y1
## 295.39
```

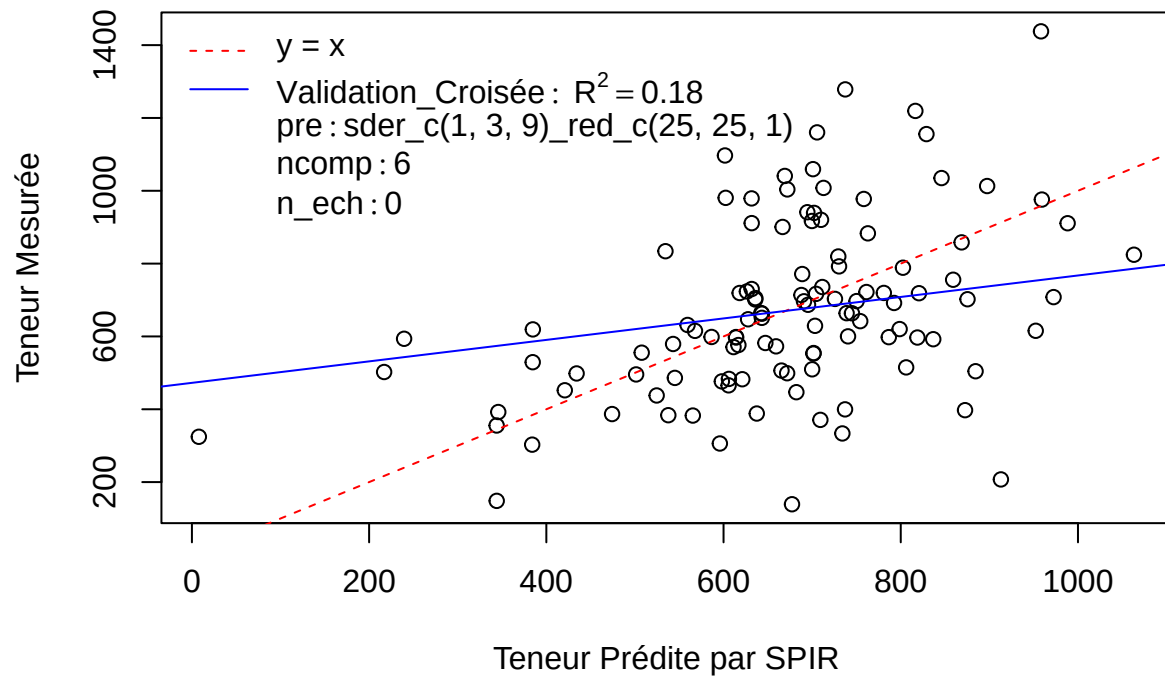


```
## [1] "TOTAL_T3"
```



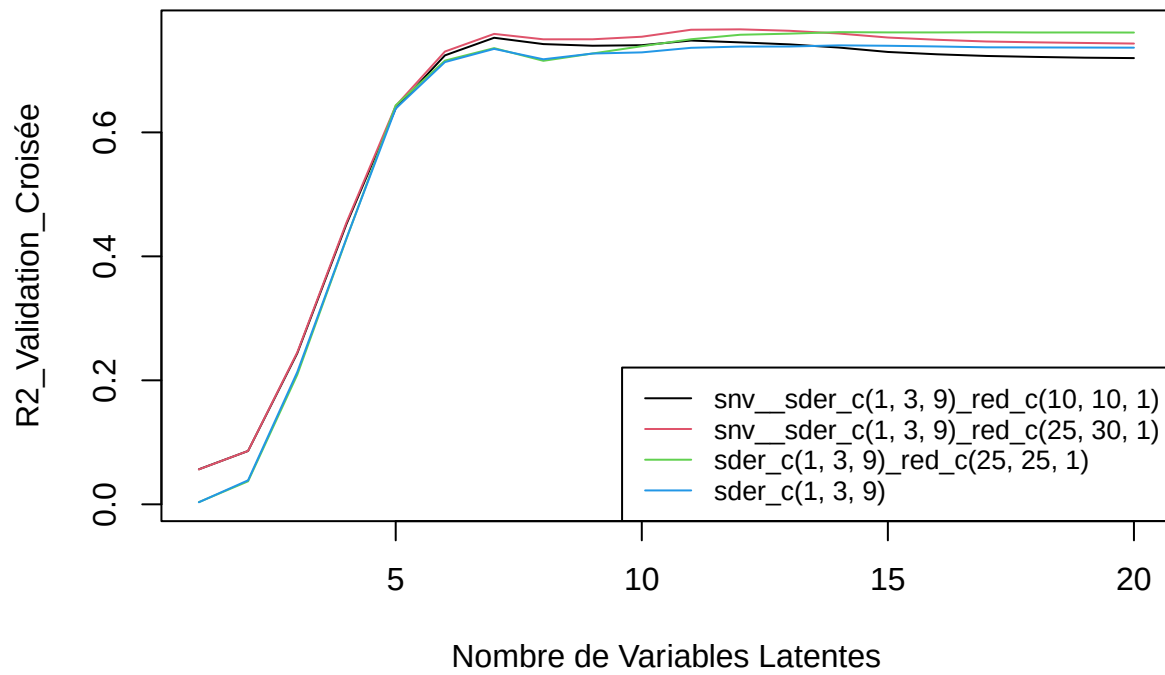
```
## [1] "sder_c(1, 3, 9)_red_c(25, 25, 1)"
## [1] "R2"
## [1] 0.00 0.00 0.03 0.08 0.16 0.18 0.16 0.14 0.12 0.13 0.13 0.13 0.13 0.14 0.14 0.14 0.14 0.14 0.14
## [1] "SEP"
##      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1      y1
## 242.21 242.07 243.37 236.62 226.11 225.42 233.94 239.06 244.03 243.43 245.16 244.64 244.04 243.34 241.41
##      y1
## 241.41
```

TOTAL_T3



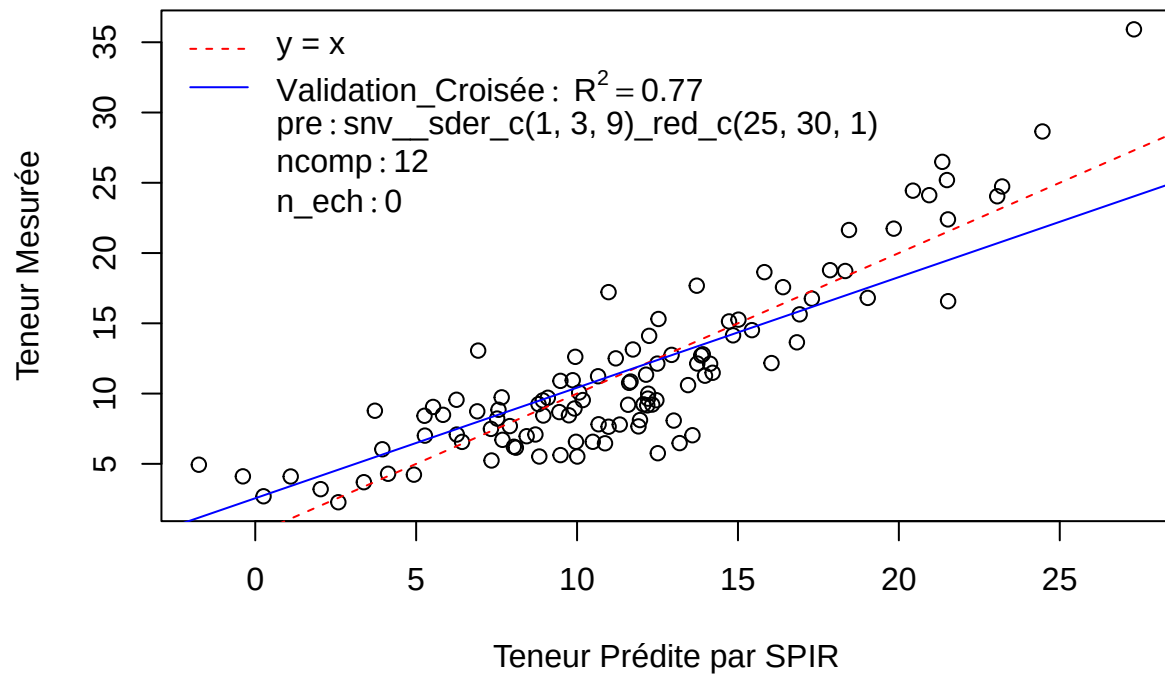
```
## [1] "FFA..."
```

FFA...



```
## [1] "snv_sder_c(1, 3, 9)_red_c(25, 30, 1)"
## [1] "R2"
## [1] 0.06 0.09 0.25 0.45 0.64 0.73 0.76 0.75 0.75 0.75 0.77 0.77 0.76 0.76 0.75 0.75 0.75 0.75 0.74
## [1] "SEP"
## y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1 y1
## 5.94 5.85 5.35 4.51 3.65 3.20 3.00 3.05 3.05 3.02 2.95 2.95 2.97 3.00 3.05 3.07 3.09 3.10 3.11 3.12
```

FFA...



Résultats ASD DIADE (défini manuellement après observation des courbes de CV)

Resultat des Cross-Validation avec les bons paramètres (PRE+nVL)

Graphique des Coefficients Beta

Coefficient beta pour 16:0, 18:1 et 18:2n9 avec Pré 1600-1750nm +
SNV + Der2, et 8VL

Prédiction de nouveaux échantillons